

Solution Requirements

Date	2 JULY 2026
Team ID	XXXXXX
Project Name	Flood Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>The system allows authorized users (meteorologists and disaster management officials) to securely log in before accessing flood prediction services.</i>	High
2	Authorization levels	<i>Different user roles are provided. Administrators manage the system, while authorized officials can enter weather data and view prediction results.</i>	High
3	External interfaces	<i>The system accepts weather data from historical datasets and supports integration with weather APIs or cloud</i>	High

		<i>platforms such as IBM Cloud.</i>	
4	Transactions processing	<i>The system receives rainfall and weather inputs, processes them using machine learning models, and instantly generates flood risk predictions.</i>	High
5	Reporting	<i>The application displays flood prediction results through dashboards and generates reports showing flood risk levels and prediction accuracy.</i>	Medium
6	Business rules, etc.	<i>The system predicts flood occurrence based on rainfall, cloud visibility, and seasonal weather conditions using trained ML algorithms (Decision Tree, Random Forest, KNN, and XGBoost).</i>	High
7	Compliance to laws or regulations	<i>The system protects user data and follows organizational policies for secure storage and responsible use of weather information.</i>	Medium
8	Other	<i>The system supports real-time prediction and future cloud deployment for remote access.</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>The system should generate flood prediction results quickly after receiving input.</i>	<i>Prediction generated within 2–3 seconds</i>
2	Scalability	<i>The application should support multiple users and increasing weather datasets without performance degradation.</i>	<i>Supports multiple concurrent users</i>
3	Security & Data Privacy	<i>User login credentials and weather data should be securely stored and protected from unauthorized access.</i>	<i>Secure authentication and encrypted data storage</i>
4	Reliability & Availability	<i>The application should remain available during emergency situations with minimal downtime.</i>	<i>99.9% system availability</i>
5	Usability & Accessibility	<i>The web interface should be simple, responsive, and easy for disaster management officials to use.</i>	<i>User-friendly interface accessible on desktop and mobile</i>
6	<i>Other</i>	<i>The system should be easy to update with new datasets and deploy on cloud platforms like IBM Cloud.</i>	<i>Supports future maintenance and cloud deployment</i>